Exit Times and Exit Distributions for Absorbing Discrete-Time Markov Chains

Theory, Derivation and Monte Carlo Validation

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Outline

1. Preliminaries: Notation and Classification of States
2. Partitioning the Transition Matrix
3. Part 1: Exit Time
4. Part 2: Exit Distribution
5. Implementation and Validation

Setup and Notation (following Durrett)

Let $(X_n)_{n\geq 0}$ be a discrete-time Markov chain on a finite state space S with transition probabilities $p(x, y) = P(X_{n+1} = y \mid X_n = x)$.

For a fixed initial state, we write

$$P_x(A) = P(A \mid X_0 = x), \qquad E_x[\,\cdot\,]$$

for the corresponding probability and expectation.

For $y \in S$, the **first hitting time of** y is

$$T_y = \min\{\, n \geq 1 : X_n = y \,\}.$$

The probability that the chain, started at y, ever returns to y is

$$\rho_{yy} = P_y(T_y < \infty).$$

Transience and Recurrence

Definition 1

A state y is **recurrent** if $\rho_{yy} = 1$ (the chain returns to y infinitely often, almost surely), and **transient** if $\rho_{yy} < 1$ (the chain visits y only finitely many times, a.s.).

Writing $x \to y$ when $\rho_{xy} = P_x(T_y < \infty) > 0$ ("x can reach y"), one shows:

- if $\rho_{xy} > 0$ but $\rho_{yx} < 1$, then x is transient;
- if x is recurrent and $x \to y$, then y is recurrent, and in fact $\rho_{yx} = 1$.

A set C is **closed** if $x \in C, y \notin C \Rightarrow p(x, y) = 0$, and **irreducible** if every pair of states in C communicates.

- A closed finite set has atleast one recurrent state.

These prove that if C is a finite closed irreducible set, then all states in C are recurrent.

Decomposition of a Finite State Space

Theorem 2 (State-space decomposition)

If S is finite, it splits as a disjoint union

$$S = T \cup R_1 \cup R_2 \cup \cdots \cup R_k,$$

where T is the set of transient states and each R_i is a closed, irreducible set of recurrent states.

Idea of proof. Take $T = \{x : \exists\, y \text{ with } x \to y \text{ but } y \not\to x\}$; every such x is transient. For any remaining state x, the set $C_x = \{y : x \to y\}$ is closed and irreducible, hence (by the recurrence lemma above) consists entirely of recurrent states; repeating this for the states not yet classified produces the recurrent classes $R_1, \ldots, R_k$.

This decomposition is exactly what lets us separate the chain into a *transient part* T that is eventually left forever, and a *recurrent part* $R = R_1 \cup \cdots \cup R_k$ that, once entered, is never left.

Block Form of P and the Fundamental Matrix

Order the states so transient states $T = \{1, \ldots, t\}$ come first and recurrent states $R = \{t+1, \ldots, t+r\}$ come second. Since R is closed, P has the block structure

$$P = \begin{pmatrix} Q & S \\ 0 & * \end{pmatrix}$$

- Q: transitions among transient states ($T \to T$).
- S: transitions from transient to recurrent states ($T \to R$).
- 0: recurrent states never return to T.

Why is $I - Q$ invertible?

Since every state in T is transient and the state space is finite, the chain eventually leaves T almost surely. Hence,

$Q^n \longrightarrow 0 \quad \Longrightarrow \quad \rho(Q) < 1$*(proof here)*, where $\rho(Q) = \max_{\lambda \in \sigma(Q)} |\lambda|$ is the spectral radius of Q.

Consequently, 1 is not an eigenvalue of Q, and therefore

$$\boxed{I - Q \text{ is invertible.}}$$

The Fundamental Matrix

Since $\rho(Q) < 1$, the matrix geometric series converges:

$$\boxed{N = (I - Q)^{-1} = I + Q + Q^2 + Q^3 + \cdots}$$

Probabilistic interpretation

Recall that $(Q^n)_{ij}$ is the probability of being at transient state j after n steps, starting from transient state i, without having left T in between. Therefore,

$$N_{ij} = \sum_{n=0}^{\infty} (Q^n)_{ij}$$

represents the expected total number of visits to transient state j, starting from transient state i, before entering R.

Thus, N is called the **fundamental matrix**.

Exit Time: Definition

Definition 3

The **exit time** from T is

$$\tau = \min\{ n \geq 0 : X_n \in R \}.$$

For $i \in T$, let $t_i = E_i[\tau]$ be the expected number of steps taken to leave T when starting at i.

Exit Time: First-Step Derivation

Fix $i \in T$. Condition on the first jump X_1:

$$t_i = E_i[\tau] = 1 + \sum_{k \in T} P(X_1 = k \mid X_0 = i)\, E_k[\tau \mid X_1 = k],$$

and $E_k[\tau \mid X_1 = k] = 0$ once $k \in R$ (the chain has already exited), while it equals t_k if $k \in T$ (strong Markov property, chain restarts at k). Hence

$$t_i = 1 + \sum_{k \in T} p(i,k)\, t_k = 1 + \sum_{k \in T} Q_{ik}\, t_k, \qquad i \in T.$$

In vector form, with $\mathbf{1}$ the all-ones vector,

$$\mathbf{t} = \mathbf{1} + Q\,\mathbf{t} \implies (I - Q)\,\mathbf{t} = \mathbf{1}.$$

Exit Time: Closed Form

Theorem 4

Let $N := (I - Q)^{-1}$ (the **fundamental matrix***). Then the vector of expected exit times is*

$$\boxed{\mathbf{t} = N\mathbf{1}}$$

Exit Distribution: Definition

Definition 5

For $i \in T$ and $j \in R$, the **exit distribution** is

$$\beta_{ij} \;=\; P_i\big(X_\tau = j\big),$$

the probability that the *first* recurrent state entered, starting from i, is exactly j.

Exit Distribution: First-Step Derivation

Fix $i \in T$, $j \in R$. Condition on X_1 exactly as before:

$$\beta_{ij} = \sum_{k \in T} P(X_1 = k \mid X_0 = i)\, P_k(X_\tau = j) + \sum_{k \in R} P(X_1 = k \mid X_0 = i)\, \mathbf{1}[k = j].$$

The first sum uses the strong Markov property: if $k \in T$, the chain restarts at k and $P_k(X_\tau = j) = \beta_{kj}$ by definition. The second sum is immediate: if the chain is already *at* j at time 1, it has exited with $X_\tau = j$; if it is at some *other* recurrent state $k \neq j$, then $X_\tau = k \neq j$, contributing 0. Hence

$$\beta_{ij} = \sum_{k \in T} Q_{ik}\, \beta_{kj} + S_{ij}, \qquad i \in T,\ j \in R.$$

Exit Distribution: Closed Form

In matrix form, with $B = (\beta_{ij})_{i \in T, j \in R}$,

$$B = QB + S \implies (I - Q)B = S.$$

We summarise as:

Theorem 6

$$\boxed{B = (I - Q)^{-1} S = NS}$$

What Is Simulated and What Is Validated

Simulated (Monte Carlo): for a chosen start $i \in T$, generate n independent trajectories of the chain, each run forward one step at a time (sampling from row i of P) until it first lands in R; record the absorbing state reached and the number of steps taken.

Computed (exact): from the same Q, S blocks, the closed-form $\mathbf{t} = N\mathbf{1}$ and $B = NS$ derived above.

Validated: empirical estimates against the exact values, using a c95% confidence interval, not a raw difference — the tolerance must scale with sample size n and the variance of exit times, which a fixed threshold cannot do.

Simulation Algorithm (per start state)

```
current = start   (repeated for n independent runs)
step = 0
while current not in R:
    step += 1
    current = sample next state from row P[current]
record: absorbed_in = current, exit_steps = step
```

Empirical estimators from n runs:

$$\hat{p}_j = \frac{\#\{\text{runs absorbed in } j\}}{n}, \qquad \bar{T}_n = \frac{1}{n}\sum_{k=1}^{n} \text{exit_steps}_k.$$

Statistical Validation: Mean Exit Time

By the CLT, with $\hat{\sigma}$ the sample standard deviation of the per-run exit times,

$$\mathrm{SE} = \frac{\hat{\sigma}}{\sqrt{n}}, \qquad \frac{\bar{T}_n - t_i}{\mathrm{SE}} \xrightarrow{d} \mathcal{N}(0,1).$$

Validation check:

$$\left| \bar{T}_n - t_i \right| \leq 1.96 \cdot \mathrm{SE} \quad \text{(theoretical value inside the empirical 95\% CI).}$$

This is a two-sided z-test at the 5% level; the tolerance widens automatically for noisier chains and tightens as n grows, which a fixed $|\text{diff}| < \varepsilon$ threshold cannot do.

Pipeline: Input File → Output File

- `input.json`: transition matrix, labels, transient/recurrent index sets, list of 5–7 starting states, number of runs n.
- For each start state: simulate n trajectories, compute $(\hat{p}_j, \bar{T}_n)$, compute $(\beta_{i\cdot}, t_i)$ from $N = (I - Q)^{-1}$.
- `output.txt`: a per-start report (empirical vs. theoretical table, 95% CI check) plus a summary table across all starts.

Running from multiple transient start states lets us check that N, t and B — computed *once* from Q, S — correctly predict every row, not just one.

Sample Result (One Start State)

Final state	Empirical	Theoretical	|Diff|
Abs (R_0)	0.4209	0.4216	0.0007
Rec (R_1)	0.2958	0.2941	0.0017
Rec (R_2)	0.2834	0.2843	0.0009

$\bar{T}_n = 1.4145$ (± 0.0046, 95% CI), $t_i = 1.4107$ $\Rightarrow$ inside CI: True.

Repeating across 5–7 start states and tabulating $\bar{T}_n$ vs. t_i side by side gives a compact, reproducible check that simulation and theory agree everywhere, not just at one starting point.

Thank You

Questions?